

Continuing SOLUTIONS_LEVEL2 with more numerical/basic MCQs (151–200):

SECTION 6: NUMERICAL MCQs – RAOULT'S LAW & COLLIGATIVE PROPERTIES (Q151–Q200)

151. The vapour pressure of pure water at 25 °C is 23.8 mm Hg. A solution of 1 mol non-volatile solute and 9 mol water has vapour pressure (assuming ideal)
- A. 23.8 mm Hg
 - B. 21.4 mm Hg
 - C. 20.0 mm Hg
 - D. 2.38 mm Hg
152. 1 mol of non-volatile solute is dissolved in 9 mol of water. Relative lowering of vapour pressure is
- A. 0.1
 - B. 0.9
 - C. 0.01
 - D. 1
153. A solution is prepared by dissolving 2 mol of non-volatile solute in 18 mol of water. Mole fraction of solute is
- A. 0.1
 - B. 0.2
 - C. 0.9
 - D. 0.1/0.9
154. Vapour pressure of benzene is lowered by 20% when a non-volatile solute is added. Mole fraction of solute is
- A. 0.2
 - B. 0.8
 - C. 0.02
 - D. 0.5
155. The boiling point of pure water is 373 K. A solution raises it to 374 K. Elevation in boiling point is
- A. 1 K

- B. 274 K
C. 373 K
D. 0.01 K
156. If K_b for water is $0.52 \text{ K kg mol}^{-1}$, elevation in boiling point for 1 molal solution is
A. 0.52 K
B. 5.2 K
C. 0.0052 K
D. 1 K
157. Depression in freezing point for 1 molal solution in water ($K_f = 1.86 \text{ K kg mol}^{-1}$) is
A. 1.86 K
B. 0.186 K
C. 0.0186 K
D. 18.6 K
158. A 0.5 molal solution (non-electrolyte) in water shows depression in freezing point of
A. 0.93 K
B. 1.86 K
C. 0.5 K
D. 0.186 K
159. The freezing point of pure water is 0°C . A solution shows freezing point -1.86°C . Its molality is ($K_f = 1.86 \text{ K kg mol}^{-1}$)
A. 0.5 m
B. 1.0 m
C. 2.0 m
D. 0.1 m
160. Osmotic pressure of 0.1 M urea solution at 300 K ($R \approx 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$) is approximately
A. 0.246 atm
B. 2.46 atm
C. 24.6 atm
D. 0.0246 atm
161. Osmotic pressure of 0.2 M glucose solution at 300 K is (approx.)
A. 0.492 atm
B. 4.92 atm

- C. 49.2 atm
- D. 0.0492 atm

162. If 0.1 M solution has osmotic pressure π at T, 0.2 M solution of same solute at same T has

- A. $\pi/2$
- B. π
- C. 2π
- D. 4π

163. 1 mol of glucose is dissolved in 1 kg water. 1 mol of NaCl is dissolved in another 1 kg water (assume complete dissociation). Which solution has higher boiling point?

- A. Glucose
- B. NaCl
- C. Both same
- D. Cannot say

164. A solution contains 0.5 mol NaCl in 1 kg water (complete dissociation). Effective molality for colligative property is ($i \approx 2$)

- A. 0.5 m
- B. 1.0 m
- C. 0.25 m
- D. 2.0 m

165. If 1 mol of acetic acid dimerizes completely in benzene, effective number of particles per initial mol is

- A. 1
- B. 0.5
- C. 2
- D. 3

166. A solution of non-electrolyte has expected $\Delta T_f = 2$ K but observed $\Delta T_f = 1$ K. This suggests

- A. Association ($i < 1$)
- B. Dissociation ($i > 1$)
- C. Ideal behaviour
- D. No solute

167. A solution of electrolyte has expected $\Delta T_f = 2$ K, observed $\Delta T_f = 4$ K. This suggests

- A. Association

- B. Dissociation
- C. $i \approx 0.5$
- D. No solute

168. If measured colligative property is twice the calculated for nonelectrolyte, van't Hoff factor i is

- A. 0.5
- B. 1
- C. 2
- D. 4

169. For AB dissociating as $AB \rightarrow A^+ + B^-$, van't Hoff factor in terms of degree of dissociation α is

- A. $1 + \alpha$
- B. $1 + 2\alpha$
- C. $1 - \alpha$
- D. $1 - \alpha/2$

170. For dimerization $2A \rightleftharpoons A_2$, van't Hoff factor in terms of α (fraction dimerized) is

- A. $1 - \alpha$
- B. $1 - \alpha/2$
- C. $1 + \alpha$
- D. $1 + 2\alpha$

171. Boiling point of a 1 molal NaCl solution (assume $i=2$, $K_b=0.52 \text{ K kg mol}^{-1}$, water) is approx.

- A. 100.52°C
- B. 101.04°C
- C. 101.5°C
- D. 102°C

172. Freezing point of 1 molal NaCl solution ($i=2$, $K_f=1.86 \text{ K kg mol}^{-1}$) is approx.

- A. -1.86°C
- B. -3.72°C
- C. -0.93°C
- D. 0°C

173. Relative lowering of vapour pressure equals

- A. Mole fraction of solute
- B. Mole fraction of solvent

C. Molality

D. Molarity

174. In a solution of 0.5 mol solute and 9.5 mol solvent, relative lowering of vapour pressure is

A. 0.5

B. 0.95

C. 0.05

D. 0.005

175. For an aqueous solution, vapour pressure is 0.98 of pure water. Mole fraction of solute is

A. 0.02

B. 0.98

C. 0.2

D. 0.002

176. Which will have higher osmotic pressure at same T?

A. 0.1 M glucose

B. 0.1 M urea

C. 0.1 M NaCl (assume complete)

D. All same

177. If two solutions A and B have same osmotic pressure at same temperature, then

A. They must have same solute

B. They must have same molarity if both non-electrolytes

C. They must have same molality

D. They are always isotonic

178. In reverse osmosis desalination, applied pressure on seawater is

A. Less than osmotic pressure

B. Equal to osmotic pressure

C. Greater than osmotic pressure

D. Irrelevant

179. Colligative property most sensitive for high-molar-mass solutes (polymers) is

A. Elevation in boiling point

B. Depression in freezing point

C. Osmotic pressure

D. Lowering of vapour pressure

180. If 1 g of solute A in 100 g water gives same ΔT_f as 2 g of solute B in 100 g water (both non-electrolytes), then
- A. Molar mass of A = molar mass of B
 - B. A has half molar mass of B
 - C. A has double molar mass of B
 - D. No relation
181. Lowering of vapour pressure is used to determine
- A. Density
 - B. Molar mass
 - C. Boiling point
 - D. Freezing point
182. Van't Hoff factor of 1.5 suggests
- A. 50% association
 - B. 50% dissociation for 1 $\rightarrow$ 2 type
 - C. 100% dissociation
 - D. No solute
183. Depression in freezing point of 0.1 m solution is 0.186 K ($K_f=1.86$). For 0.1 m solution of another solute same solvent, $\Delta T_f = 0.372$ K. Van't Hoff factor is
- A. 1
 - B. 2
 - C. 0.5
 - D. 4
184. Osmotic pressure is preferred for molar mass determination because
- A. It can be measured at room T and very dilute solutions
 - B. It requires high T
 - C. It is independent of concentration
 - D. It is zero for all
185. Ideal semipermeable membrane for osmosis allows
- A. Only solute
 - B. Only solvent
 - C. Both equally
 - D. Neither

186. A solution containing 1 mol glucose in 1 L and another containing 1 mol urea in 2 L at same T
- A. Are isotonic
 - B. Have same osmotic pressure
 - C. First has double osmotic pressure of second
 - D. Second has double osmotic pressure
187. When two solutions are separated by semipermeable membrane and there is no net osmosis, they are
- A. Hypotonic
 - B. Hypertonic
 - C. Isotonic
 - D. Supersaturated
188. If freezing point of solution is -0.93°C , its elevation in boiling point for same solvent and molality ($K_b=0.52$, $K_f=1.86$) is approx.
- A. 0.26°C
 - B. 0.52°C
 - C. 0.93°C
 - D. 1.86°C
189. For an aqueous solution of non-volatile solute, which of the following is true?
- A. T_b increases, T_f increases
 - B. T_b increases, T_f decreases
 - C. T_b decreases, T_f decreases
 - D. Both unchanged
190. If 1 mol of Na_2SO_4 dissociates completely into three ions, theoretical van't Hoff factor is
- A. 2
 - B. 3
 - C. 1
 - D. 0.5
191. Association of solute molecules in solution leads to
- A. Higher than expected colligative properties
 - B. Lower than expected colligative properties
 - C. No change
 - D. Random change

192. Dissociation of solute molecules in solution leads to
- A. Higher than expected colligative properties
 - B. Lower than expected colligative properties
 - C. No change
 - D. Random change
193. The colligative properties for 0.1 m urea, 0.1 m glucose, 0.1 m sucrose (all non-electrolytes) are
- A. All different
 - B. All same
 - C. Proportional to molar mass
 - D. Proportional to density
194. When a non-volatile solute is added to pure solvent
- A. Vapour pressure increases
 - B. Boiling point increases
 - C. Freezing point increases
 - D. Osmotic pressure decreases
195. Colligative properties for solutions at same T compare directly with
- A. Number of moles of solute present
 - B. Mass of solute present
 - C. Volume of solution
 - D. Type of solute
196. If osmotic pressure of solution doubles, at constant T and solvent, concentration
- A. Doubles
 - B. Halves
 - C. Becomes four times
 - D. Becomes one-fourth
197. If solution A has greater depression in freezing point than solution B, both in same solvent
- A. A has smaller number of solute particles
 - B. A has more solute particles
 - C. Both have equal number
 - D. No conclusion
198. Vapour pressure of pure solvent is 300 torr. Solution with non-volatile solute has vapour pressure 270 torr. Relative lowering is

- A. 0.1
- B. 0.9
- C. 0.3
- D. 0.27

199. Colligative properties are particularly useful to find

- A. Empirical formula
- B. Molar mass of unknown non-volatile solute
- C. Atomic mass
- D. Structure

200. Total number of MCQs in SOLUTIONS_LEVEL2 set is

- A. 100
- B. 150
- C. 200
- D. 250